Chapter 1

Number System

A system in which we study different types of numbers, their relationship and rules governing them is called a number system. In the Hindu-Arabic system, we use the symbols $0, 1, 2, 3, 4, 5, 6, 7, 8$ and 9. These symbols are called digits. Of these ten digits, 0 is called an **insignificant digit** whereas the others are called **significant digits**.

1.1 Numerals

A mathematical symbol representing a number in a systematic manner is called a numeral represented by a set of digits.

How to Write a Number

To write a number, we put digits from right to left in places designated as units, tens, hundreds, thousands, ten thousands, lakhs, ten lakhs, crores, and ten crores. Let us see how the number 308761436 is denoted. It is read as

Ten crores	**Crores**	**Ten lakhs**	**Lakhs**	**Ten thousands**	**Thousands**
10^8	10^7	10^6	10^5	10^4	10^3
$\downarrow$	$\downarrow$	$\downarrow$	$\downarrow$	$\downarrow$	$\downarrow$
3	0	8	7	6	1

Hundreds	**Tens**	**Units**
10^2	10^1	10^0
$\downarrow$	$\downarrow$	$\downarrow$
4	3	6

Thirty crore eighty seven lakh sixty one thousand four hundred and thirty six.

1.2 Face Value and Place Value

Face Value

In a numeral, the face value of a digit is the value of the digit itself irrespective of its place in the numeral. For example, in the numeral 486729, the face value of 8 is 8, the face value of 7 is 7, the face value of 6 is 6, the face value of 4 is 4, and so on.

Place Value (or Local Value)

In a numeral, the place value of a digit changes according to the change of its place. Look at the following to get the idea of place value of digits in 72843016.

Crores 7	$\longrightarrow$	Place Value of 7	$\longrightarrow$	$7 \times 10000000 = 70000000$
Ten Lakhs 2	$\longrightarrow$	Place Value of 2	$\longrightarrow$	$2 \times 1000000 = 2000000$
Lakhs 8	$\longrightarrow$	Place Value of 8	$\longrightarrow$	$8 \times 100000 = 800000$
Ten Thousands 4	$\longrightarrow$	Place Value of 4	$\longrightarrow$	$4 \times 10000 = 40000$
Thousands 3	$\longrightarrow$	Place Value of 3	$\longrightarrow$	$3 \times 1000 = 3000$
Hundreds 0	$\longrightarrow$	Place Value of 0	$\longrightarrow$	$0 \times 100 = 0$
Tens 1	$\longrightarrow$	Place Value of 1	$\longrightarrow$	$1 \times 10 = 10$
Ones 6	$\longrightarrow$	Place Value of 6	$\longrightarrow$	$6 \times 1 = 6$

It is clear from the above presentation that to obtain the place value of a digit in a numeral, we multiply the digit with the value of its place in the given numeral.

1.3 Types of Numbers

1.3.1 Natural Numbers ($\mathbb{N}$)

Natural numbers are counting numbers. They are denoted by $\mathbb{N}$. The natural numbers are $\mathbb{N} = \{1, 2, 3, \dots\}$.

- ✓ All natural numbers are positive.
- ✓ Zero is not a natural number. Thus, 1 is the smallest natural number.

Even Numbers and Odd Numbers

- A counting number which is **divisible by 2**, is called an even number.
  For example: $2, 4, 6, 8, 10, 12, \dots$ etc.

  The unit's place of every even number will be 0, 2, 4, 6 or 8.
- A counting number which is **not divisible by 2**, is known as an odd number.
  For example: $1, 3, 5, 7, 9, 11, 13, 15, 17, 19, \dots$ etc.

  The unit's place of every odd number will be 1, 3, 5, 7 or 9.

Prime Numbers and Composite Numbers

A counting number is called a prime number when it is exactly divisible by, 1 and itself.
For example: 2, 3, 5, 7, 11, 13, ... etc.

- 2 is the only even number which is prime.
- A prime number is always greater than 1.
- 1 is not a prime number. Thus, the smallest odd prime number is 3.
- Every prime number greater than 3 can be represented by $6n + 1$, where n is integer.

Note: How to test a number is Prime or Not

If p is the given number, then

1. Find smallest whole number x such that $x > \sqrt{p}$

2. Take all prime numbers less than or equal to x
3. If none of these divide p exactly, then p is prime. Otherwise p in not prime.

For example: Let $p = 193$

Clearly $14 > \sqrt{193}$

Prime numbers upto 14 are $2, 3, 5, 7, 11, 13$.

None of these divides 193 exactly.

Hence, 193 is a prime number.

Composite numbers are non-prime natural numbers. They must have at least one factor apart from 1 and itself.
For example: 4, 6, 8, 9, *etc*.

- Composite numbers can be both odd and even.
- 1 is neither a prime number nor composite number.

Coprimes

Two natural numbers are said to be coprimes, if their HCF is 1.
For example: $(7, 9)$, $(15, 16)$

- Coprime numbers may or may not be prime.

1.3.2 Whole Numbers ($\mathbb{W}$)

All natural numbers and zero form the set of whole numbers. Whole numbers are denoted by $\mathbb{W}$. For example $\mathbb{W} = \{0, 1, 2, 3, ...\}$

- Zero is the smallest whole number.

Whole numbers are also called as non-negative integers.

1.3.3 Integers ($\mathbb{I}$ or $\mathbb{Z}$)

Whole numbers and negative numbers form the set of integers. They are denoted by $\mathbb{I}$. *For example:* $\mathbb{I} = \{..., -4, -3, -2, -1, 0, 1, 2, 3, 4, ...\}$.
Integers are of two types.

Positive Integers and Negative Integers

Positive Integers: Natural numbers are called as positive integers. They are denoted by $\mathbb{I}+$. *For example:* $\mathbb{I}+ = \{1, 2, 3, 4, ...\}$.
Negative Integers: Negative natural numbers are called as negative integers. They are denoted by $\mathbb{I}-$. *For example:* $\mathbb{I} = \{-1, -2, -3, -4, ...\}$.

- '0' is neither +ve nor -ve integer.

1.3.4 Rational Numbers ($\mathbb{Q}$)

A number that can be expressed as p/q, where p and q are integers and $q \neq 0$ is called a rational number.
For example: $\frac{3}{5}, \frac{7}{9}, \frac{8}{9}, 2 = \frac{2}{1}, 0 = \frac{0}{1}, -5 = \frac{-5}{1}...$

1.3.5 Irrational Numbers ($\mathbb{Q}^*$)

The numbers that cannot be expressed in the form of p/q, where p and q are integers and $q \neq 0$ are called irrational numbers.
For example: $\sqrt{2}, \sqrt{3}, e, \pi/2,\ etc.$

- π is an irrational number as $22/7$ is not the actual value of π but it is its nearest value.
- Non-periodic infinite decimal fractions are called as irrational number.

1.3.6 Real Numbers ($\mathbb{R}$)

Real numbers include rational and irrational numbers both.
For example: $0, -1, 5, \frac{3}{5}, \sqrt{7}, \pi/2.$
Real numbers are denoted by $\mathbb{R}$.

1.4 Operations on Numbers

Addition

When two or more numbers are combined together, then it is called addition. Addition is denoted by $'+'$ sign. *For example:* $24 + 23 + 26 = 73$

Subtraction

When one or more numbers are taken out from a another number, then it is called subtraction. Subtraction is denoted by $'-'$ sign.
For example: $100 - 4 - 13 = 100 - 17 = 83$

Multiplication

When $'a'$ is multiplied by $'b'$, then $'a'$ is added $'b'$ times or $'b'$ is added $'a'$ times. It is denoted by $'\times'$. Let us see the following operation on Multiplication.
If $a = 2$ and $b = 4$, then $2x4 = 8$ or $(2 + 2 + 2 + 2) = 8$
Here, $'a'$ is added $'b'$ times or in other words 2 is added 4 times.
Similarly, $4x2 = 8$ or $(4 + 4) = 8$
In this case, $'b'$ is added $'a'$ times or in other words 4 is added 2 times

Division

When D and d are two numbers, then $\frac{D}{d}$ is called the operation of division, where D is the dividend and d is the divisor. A number which tells how many times a divisor(d) exists in dividend(D) is called the quotient(Q). If dividend(D) is not a multiple of divisor(d) , then D is not exactly divisible by d and in this case remainder(R) is obtained.
Let us see the following operation of division.

Let $D = 17$ and $d = 3$

Then, $\frac{D}{d} = \frac{17}{3} = 5\frac{2}{3}$

Here, $5 =$ Quotient(Q),
$3 =$ Divisor(d)
$2 =$ Remainder(R)

We See

$$3 \text{ (Divisor)} \times 5 \text{ (Quotient)} + 2 \text{ (Remainder)} = 17 \text{ (Dividend)}$$

Hence, We can write the formula

$$\text{Dividend} = (\text{Divisor} \times \text{Quotient}) + \text{Remainder}$$

1.5 Divisibility Tests

- **Divisibility by 2:** When the last digit of a number is either 0 or even, then the number is divisible by 2. *For example:* 12, 86, 472, 520, 1000 etc., are divisible by 2.
- **Divisibility by 3:** When the sum of the digits of a number is divisible by 3, then the number is divisible by 3. *For example:*
  (a) 1233 $\implies$ 1 + 2 + 3 + 3 = 9, which is divisible by 3, So 1233 is divisible by 3.
  (b) 156 $\implies$ 1 + 5 + 6 = 12, which is divisible by 3, So 156 must be divisible by 3.
- **Divisibility by 4:** When the number made by last two-digits of a number is divisible by 4, then that particular number is divisible by 4. Apart from this, the number having two or more zeroes at the end, is also divisible by 4.
  For example:
  (a) 6428 is divisible by 4 as the number made by its last two digits *i.e.,* 28 is divisible by 4.
  (b) The numbers 4300, 153000, 9530000 etc., are divisible by 4 as they have two or more zeroes at the end.
- **Divisibility by 5:** Numbers having 0 or 5 at the end are divisible by 5.
  For example: 45, 4350, 135, 14850 etc., are divisible by 5 as they have 0 or 5 at the end.
- **Divisibility by 6:** When a number is divisible by both 3 and 2, then that particular number is divisible by 6 also.*For example:* 18, 36, 720, 1440 etc., are divisible by 6 as they are divisible by both 3 and 2.
- **Divisibility by 7:** A number is divisible by 7 when the difference between twice the digit at ones place and the number formed by other digits is either zero or a multiple of 7.
  For example: 658 is divisible by 7 because $65 - 2 \times 8 = 65 - 16 = 49$. As 49 is divisible by 7, the number 658 is also divisible by 7.
- **Divisibility by 8:** When the number made by last three digits of a number is divisible by 8, then the number is also divisible by 8. Apart from this, if the

last three or more digits of a number are zeroes, then the number is divisible by 8. *For example*

(a) 2256, As 256 (the last three digits of 2256) is divisible by 8, therefore 2256 is also divisible by 8.

(b) 4362000, As 4362000 has three zeroes at the end. Therefore it will definitely divisible by 8.

❒ **Divisibility by 9:** When the sum of all the digits of a number is divisible by 9, then the number is also divisible by 9. *For example:*

(a) 936819 $\implies$ 9+3 + 6 + 8 + 1 + 9= 36 which is divisible by 9. Therefore, 936819 is also divisible by 9.

(b) 4356 $\implies$ 4 + 3 + 5 + 6 = 18 which is divisible by 9. Therefore, 4356 is also divisible by 9.

❒ **Divisibility by 10:** When a number ends with zero, then it is divisible by 10.
For example: 20, 40, 150, 123450, 478970 etc., are divisible by 10 as these all end with zero.

❒ **Divisibility by 11:** When the sums of digits at odd and even places are equal or differ by a number divisible by 11, then the number is also divisible by 11.
For example:

(a) For 2865423, Sum of digits at odd places (A) = 2 + 6 + 4 + 3 = 15, Sum of digits at even places (B) = 8 + 5 + 2 = 15 $\implies$ $A = B$. Hence, 2865423 is divisible by 11.

(b) For, 217382 Sum of digits at odd places (A) = 2 + 7 + 8 = 17, Sum of digits at even places (B) = 1 + 3 + 2 = 6 $\implies$ $A - B = 17 - 6 = 11$ Clearly, 217382 is divisible by 11.

❒ **Divisibility by 12:** A number which is divisible by both 4 and 3 is also divisible by 12. *For example:* 2244 is divisible by both 3 and 4. Therefore, it is divisible by 12 also.

❒ **Divisibility by 14:** A number which is divisible by both 7 and 2 is also divisible by 14. *For example:* 1232 is divisible by both 7 and 2. Therefore, it is divisible by 14 also.

- **Divisibility by 15:** A number which is divisible by both 5 and 3 is divisible by 15 also. *For example:* 1275 is divisible by both 5 and 3. Therefore, it is divisible by 15 also.
- **Divisibility by 16:** A number is divisible by 16 when the number made by its last 4-digits is divisible by 16. *For example:* 126304 is divisible by 16 as the number made by its last 4-digits i.e., 6304 is divisible by 16.
- **Divisibility by 18:** A number is divisible by 18 when it is even and divisible by 9. *For example:* 936198 is divisible by 18 as it is even and divisible by 9.
- **Divisibility by 25:** A number is divisible by 25 when its last 2-digits are either zero or divisible by 25. *For example:* 500, 1275, 13550 are divisible by 25 as last 2-digits of these numbers are either zero or divisible by 25.
- **Divisibility by 125:** A number is divisible by 125 when the number made by its last 3-digits is divisible by 125. *For example:* 630125 is divisible by 125 as the number made by its last 3-digits are divisible by 125.

To Find a Number Completely Divisible by Given Number

Consider a number x which is when divided by d, gives a quotient q and leaves a remainder r. Then,

$$d\overline{)\begin{array}{r}q\\ x\\ \hline r\end{array}}$$

To find the number which is completely divisible by d such that remainder r is zero, follows the example given below.

Ex. 1: Find the number, which on (i) addition (ii) subtraction from the number 5029 is completely divisible by 17.

Sol. Dividing 5029 by 17 we get remainder = 14

$$
\begin{array}{r}
295 \\
17\overline{)5029} \\
34 \\
\hline
162 \\
153 \\
\hline
99 \\
85 \\
\hline
14
\end{array}
$$

(i) The minimum number on adding of which the given number is completely divisible by 17

= Divisor − Remainder = 17 − 14 = 3.

$\therefore$ The number is 5029 + 3 = 5032

(ii) The minimum number on subtraction of which the given number is completely divisible by 17

= Remainder = 14.

$\therefore$ The number is 5029 − 14 = 5015.

Unit's Place of an Expression

Given expression can be of following two types

When Number is Given in the form of Product of Number

To find the units digit in the product of two or more number we take units digit of every numbers and then multiply them. Then, the unit digit of the resultant product is the units digit of the product of original numbers.

For example: $207 \times 781 \times 39 \times 94$

Taking units digit of every number and then multiplying them

$= 7 \times 1 \times 9 \times 4 = 7 \times 36$ [taking units place digit]

Again, taking units digit and then multiplying $= 7 \times 6 = 42$

$\therefore$ Units digit for $207 \times 781 \times 39 \times 94$ is 2.

When Number is Given in the form of Index

1. If the unit's digit number are 0,1, 5 or 6, then the resultant unit's digit remains same.*For example:*
   - ➢ $(576)^{1151}$ – its units digit is 6.
   - ➢ $(155)^{120}$, its unit digit is 5.
   - ➢ $(191)^{19}$, its unit digit is 1.
   - ➢ $(900)^{51}$, its unit digit is 0.
2. If units place is 2, then the power of the number is first divided by 4 and there after represented in the form of 2 .
   For example: $(572)^{443}$

   Since units digit is 2, Dividing the power of number by 4, we have

   $443 \div 4 = 110$with remainder3

   $i.e, \quad 443 = 4 \times 110 + 3$

   $\therefore \quad 2^{443} = 2^{4\times110} \times 2^3 = (2^4)^{110} \times 2^3 = 2^4 \times 2^3$

   Since $2^4 = 16$, by (i) $(2^4)^{110} = 16^{110}$ is same as 2^4

   $\Longrightarrow \quad 16 \times 8 = 128$ taking units place digit.

   $\therefore$ Units place is 8.

3. In the same way, if units place digit are 4 or 8, then units digit for 4^4 and 8^4 are 6.
   For example: $(124)^{372}$

   Taking units place digit = $4^{372} = 4^{4\times93} = (4^4)^{93}$

   Since $4^4 = 256$ ends in 6, by (i) the unit place of the$(4^4)^{93}$ and 4^4 are the same.

   $\therefore$ Unit's place digit = 6.

4. If units digit is 3 or 7, then units digit for 3^4 and 7^4 is 1.
   For example: $(2467)^{153}$

   Taking units place = $7^{153} = 7^{4\times38+1} = (7^4)^{38} \times 7$

   Since, Units place of $7^4 = 1$.

   $\therefore$ Unit place of the given number = Unit place digit of $1^{38} \times 7 = 7$

5. If units place is 9 and if the power of 9 is even, then units digit will be 1 and if the power of 9 is odd, then units digit will be 9.
   For example:

   For $(539)^{140}$

   Since, power is even for unit's digit 9.

   Units digit for expansion is 1

   For $(539)^{141}$

   Since, power is odd for unit's digit9

   Unit digit for expansion is 9.

Basic Number Theory

- Square of every even number is an even number while square of every odd number is an odd number.
- A number obtained by squaring a number does not have 2, 3 ,7 or 8 at its unit place.
- There are 15 prime numbers between 1 and 50 and 10 prime numbers between 50 and 100.
- If p divides q and r, then p divides their sum and difference also.
  For examples, 4 divides 12 and 20, then 20 + 12 = 32 and 20 -12 = 8 are also divisible by 4.
- For any natural number n, (n - n) is divisible by 6.
- The product of three consecutive natural numbers is always divisible by 6.
- $(x^m - a^m)$ is divisible by $(x - a)$ for <u>all values</u> of m.
- $(x^m - a^m)$ is divisible by $(x + a)$ for <u>even values</u> of m.
- $(x^m + a^m)$ is divisible by $(x + a)$ for <u>odd values of</u> m.
- Number of prime factors of $a^p b^q c^r d^s$ is $p + q + r + s$, where a, b, c and d are prime numbers.

◆ Some Basic Summations

- Sum of first n natural numbers
  $1 + 2 + 3 + ...n = \frac{n(n+1)}{2}$.
- Sum of first n odd numbers
  $1 + 3 + 5 + ... + (2n - 1) = n^2$.
- Sum of first n even numbers
  $2 + 4 + 6 + ... + (2n - 2) = n(n + 1)$.
- Sum of squares of first n natural numbers
  $1^2 + 2^2 + 3^2 + ... + n^2 = \frac{n(n+1)(2n+1)}{6}$.
- Sum of cubes of first n natural numbers
  $1^3 + 2^3 + 3^3 + ... + n^3 = \left[\frac{n(n+1)}{2}\right]^2$